IMA EDUCATIONAL Case Journal | Case Study

ima® The Association of Accountants and Financial Professionals in Business

ISSN 1940-204X

# Target Costing Decisions at BMA AG*

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## INTRODUCTION

Tim Lahn is a young management accountant at the German car company BMA AG. He works in a large management accounting department in product development. His specific role is to help project teams manage costs when developing a new car model and to ensure that the finished design will allow BMA to make its target profit. To fulfill his role, Tim and his colleagues rely heavily on target costing. They develop a business case for a new car model, which includes targets for the sales price, sales volumes, and costs. The business case also specifies attributes for the functionality and performance of the car. Tim and the entire project team review the design of the new car as it progresses. Does it meet the targets, not only regarding costs, but also for all the other important attributes? Which actions need to be taken? For this case, you will fill the role of Tim or another member of the project team to help ensure the success of BMA's 3C4 project. You will also be involved in a task force for improving BMA's target-costing method. In this case, you will be focused on decision-making and target-costing concepts, and you will not be preparing detailed target-costing calculations.

The importance of target costing for BMA during product development is easy to see. Early in the product lifecycle, BMA has many more degrees of freedom to make decisions that significantly affect costs. Once a new car model has been launched, much less can be changed to reduce costs. For example, cost reduction through materials substitution would be much more difficult, as this would require other design elements of the car to be changed as well, with complex implications for production processes and suppliers.

After presenting more information about BMA, Part I of the case offers details about BMA's current 3C4 project, which is the focus of the next meeting of the 3C4 project team. Tim needs to prepare for that meeting. Part II of the case takes place about two years later and includes information about a newly formed task force for changing BMA's target-costing system, which Tim is heading. Tim needs to prepare for the first meeting of this task force. The discussions in class emulate these meetings of the 3C4 project team and the task force. The case is rather long because it provides a realistic description of target costing in practice.

## BMA'S MARKET, VALUE CHAIN, AND COST STRUCTURE

BMA AG offers car models in a range of sizes, from mini-compact to large cars and SUVs, or in the European classification, cars in the A-F, S, and J segments. Most of BMA's car models come in various vehicle types, such as four-door sedan, five-door station wagon, two-door coupe, SUV, and convertible. The company's head office is located in Southern Germany, with production facilities located in Germany and all over the world. The network of suppliers is crucial, as more than 75% of the manufacturing costs consisted of purchased parts.

**Sales market.** BMA's sales market was global. Sales prices and car specifications depended on consumer preferences and

*The case has been developed solely for class discussion and is based on interviews and other information obtained from various car companies. While the case realistically highlights the intricacies of target costing at car companies, specific examples and numbers have been disguised and are partly fictitious.

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competition in a local market. For example, competition is very tough in the United States, and prices are considerably lower in the United States than in Europe. Another difference is that in Europe, customers usually specify the car they want and then wait weeks or months before it is delivered after being assembled to order. In the United States, customers buy a fully loaded car that the dealer has in inventory. In China, customers usually buy the car from the dealer's inventory; models with extra-long wheelbases, as well as softer upholstery and suspension, are popular. Clearly, markets have their own characteristics.

**Sales prices.** Sales prices of all the cars of one model vary enormously. As mentioned previously, a car model usually comes in various vehicle types, which could be equipped with various types of engines, such as diesel, gasoline, or hybrid. A customer could choose from many more options, such as those for communication, navigation, entertainment, seat materials, headlights, all-wheel drive, manual or automatic transmission, types of suspension, glass roofs, rims, and paint. Furthermore, a car model could be sold in many different countries, where market prices differ considerably. Moreover, sales prices are dynamic, and the price in the first year after a new model has been launched differs from the price after, for example, three years. As a car model gets older, more special offers are needed in the form of lower prices and/or extra options for the same price.

**Value chain.** BMA's value chain is rather complex. Dealers operate in more than 100 countries, and manufacturers have more than 20 factories in more than 15 countries. The importing companies are sometimes owned by BMA, but others are independently owned. Cars are transported to the sales countries by trucks, trains, and ships. Some of the manufacturing factories are joint ventures with local companies, and others conduct only the simpler assembly activities based on "completely knocked down" kits. Dozens of first-tier suppliers deliver parts to BMA's factories, and these suppliers coordinate operations with thousands of second- and third-tier suppliers all over the world.

**Cost structure.** The cost structure of the value chain is quite intricate for target costing. BMA needs to consider local costs in each of its markets: the margins of importers and dealers, warranty, sales, marketing, distribution, import duties, and local value-added sales taxes. Some countries have considerable additional taxes beyond the normal value-added sales tax, which applies specifically to cars and depends on the price and technical characteristics of the car. For example, the final consumer price in Denmark could be more than twice the price before taxes. Furthermore, exchange rates entered into the equation, because many importing companies pay BMA in a currency other than the euro; likewise, BMA pays suppliers in various currencies. All these factors have an impact on the net revenues of the car—the sales price that BMA receives in Germany is typically about 60% to 70% of the average final sales price of a car, depending on the model. Because sales prices decrease over time, most costs have to be reduced as well during the lifecycle of a car model.

**Initial investments.** BMA's initial investments for developing a new car model could easily amount to €1 billion. These consisted of research and development (R&D) expenses such as for engineers and other staff, laboratories, equipment, tests, prototypes, externally hired engineers and other staff, and payments to suppliers for their R&D efforts. BMA also incurs significant development costs for conducting a product upgrade about halfway into a model's lifecycle. Furthermore, BMA makes model-specific investments in tools (such as pressing tools, tools in the body shop, containers/cartridges in assembly) and other manufacturing equipment specifically required for a particular car model. Most investments are made before the start of production but are also made later for the product upgrade. Some investments in manufacturing assets are paid for and owned by BMA but are actually placed at suppliers' sites.

**Nonmanufacturing costs.** BMA incurs many nonmanufacturing costs for a particular car model. Besides the costs for product development, costs for marketing, distribution, sales, and warranty are relevant. Marketing costs encompass BMA's marketing activities for a new model or model upgrade, such as producing and airing expensive commercials, producing brochures in many different languages, and creating the websites for the new model. Although most sales costs are incurred by the importers (BMA fully or partially owns some of these), BMA also has a host of different arrangements for contributing to local sales costs, such as product advertising and incentives and rebates for customers and dealers. Warranty costs depend on the local conditions for how cars are used and how much they wear, as well as customer expectations about leniency of dealers when it comes to warranty issues. Overall, these direct nonmanufacturing costs tend to range from 20% to 35% of the total costs, depending on the market segment for a car model. Manufacturing costs could be direct or indirect in relation to a particular car model.

**Indirect manufacturing costs.** Almost all of BMA's production sites produce multiple car models, which share assets such as manufacturing equipment (for example, the presses for producing the metal parts of a car body), assembly lines (a mix of different car models could be assembled on one line), and buildings. Support departments such as inbound logistics and quality management also work for several car models.

**Direct manufacturing costs.** Model-specific manufacturing assets cause direct, fixed manufacturing costs, but most direct manufacturing costs are variable. The variable manufacturing costs consist mainly of parts purchased from suppliers and direct manufacturing labor. As a car entails thousands of different components, parts, and materials, many detailed target costs provide the starting point for price negotiations with suppliers. BMA buys some major components, such as transmissions, as complete units from specialized suppliers. In other cases, BMA buys smaller parts and puts together a component itself. For example, BMA produces its own engines, and the same types of engines are used for various car models. BMA also produces some parts from raw materials, such as many of the pressed metal parts for the car body.

**Corporate overhead.** Finally, BMA's cost structure includes corporate overhead, such as financing costs and costs for central departments such as basic R&D, corporate communication, finance, and legal.

## PART 1: TARGET COSTING THE NEW 3C4 MODEL

BMA is currently in the middle of developing a new generation of its model 3C4, which accounts for its largest sales volume. Tim Lahn is part of the 3C4 development project. The main vehicle types of the 3C4 family are the four-door sedan and the five-door station wagon. Besides offering various types of conventional diesel and gasoline engines, BMA also introduces a plug-in hybrid and a 100% electric drive for the new generation of the 3C4 model.

Development of a new car takes around four years, and BMA applies target costing to manage the cost of the new 3C4 model during product development. Development consists of two phases: the concept phase for product definition and the realization phase for product development. Details of BMA's product development process are in Appendix A. The 3C4 project was now halfway into the realization phase, close to the procurement release milestone.

The next meeting of the 3C4 project team is taking place in two days, and Tim needs to prepare for this meeting. He recollects what had happened so far in the concept phase and the realization phase of the new 3C4 model.

### TARGET COSTING IN THE 3C4 CONCEPT PHASE

Tim believed that the concept phase had been the crucial time for target costing, when several decisions with a major impact on costs had been made. That early, it was not yet possible for him to model the full complexity of all the different countries, customer-specific configurations, price erosion, supply chain costs, and so on, as described previously. In the concept phase, cost modeling was still approximate but, at the same time, was central to guiding key technical decisions.

Tim had started target costing for the 3C4 by defining three so-called "reporting models" for Germany, the United States, and China (see Table 1 for an example). For each model, specifications were defined, sales prices and volumes were set, and the required profit margin was determined. Other assumptions were also made, such as for production locations and exchange rates. Target costing during concept design was based on in-depth analyses of these three reporting models.

#### Applying the Target Costing Logic and Finding Cost Reductions

The logic of target costing during the concept phase is shown in Exhibit 1. The requirements for the new 3C4 model were described by the marketing function in a product profile document. The document defined properties of the car for which customers had certain expectation levels, such as acceleration, maximum speed, fuel economy, connectivity, $CO_2$ and NOx emissions, types of transmissions, engines, and drive systems (two-wheel, all-wheel), interior noise, upholstery, and safety. Some properties were qualitatively described, such as for the driving experience ("agile handling") or the design ("dynamic, light"). Some requirements were based on regulation (existing or anticipated). The inputs for determining the requirements were diverse, such as analysis of the strengths and weaknesses of the current 3C4 model, research about competitors' future cars in the same market segment as the 3C4, as well as feedback and ideas from customers, dealers, importers, and BMA employees.[1]

Guided by these requirements for the new 3C4 model, simultaneous engineering (SE) teams made technical proposals for the design of the car, and cost engineers made cost estimates. These activities involved the specialized knowledge and experience of dozens of experts. In many cases, the initial proposals had been too expensive and exceeded the target cost, so other technical solutions had to be found (see examples in Appendix B). Cheaper solutions, however, still need to result in a car that satisfied all the requirements for the new 3C4 model and enabled

achievement of the targeted sales prices and sales volumes. Tim is well aware that ignoring some requirements would always be possible to reduce costs, but then the car could not be sold at the targeted price or volume. On the other hand, it would always be possible to make a great car that exceeded the requirements and would be sold in a large volume and at the targeted price; but then the allowable costs would be exceeded, and the project would not achieve its financial goals. The sales price, sales volume, and profit margin targets were the corner points for all decisions about costs and technical concepts. Allowing extra costs in one area meant costs needed to be saved in another area.

Another lever for cost reduction was global scouring and looking for other suppliers who could produce particular parts at a lower cost. BMA was buying more and more parts from suppliers in China. If the initial assessment of a potential supplier indicated the company did not qualify, BMA sometimes invested in supplier development by sending out teams to help the supplier improve.

Early in the concept phase, the production location had also been discussed as a way to reduce costs. The current 3C4 model was built in Germany. Cost pressures provoked a debate about moving the production location to Poland, which was estimated to save around €205 per unit. But concerns about employment at the main location of BMA in Germany quickly led to a board of directors decision about how changing the production location was not acceptable for BMA's highest-volume model.

Later in the concept phase, Tim had participated in an exercise for finding cost-reduction opportunities. BMA had deconstructed two cars of its main competitors and laid out 200 parts of those cars on tables across a large room. Next to each part, an estimate of its cost was shown, as well as the design of that same part for BMA's new 3C4 model with its estimated cost. The BMA board of directors had spent the entire day discussing technical differences and similarities of parts and reasons for cost differences. This deconstruction had elicited numerous ideas for how costs could be reduced and had yielded insights into which cost targets were realistic.

### Finalizing the Cost Targets

At the end of the concept phase, the managers responsible for the major areas and cost categories of the new 3C4 model had to state whether they were willing to commit to their cost targets. After intense discussions, negotiations, and some target adjustments, the technical concept was finalized, the specific targets and overall business case became the formal goals, and the project continued. In fact, it had to be continued, because the 3C4 was a central model in BMA's product portfolio. But Tim also knew that some projects, especially for derivative models, had been cancelled at the end of the concept phase once it had become clear that the required profitability targets could not be achieved. A recent example was the concept for a convertible version of one of BMA's SUVs.

## TARGET COSTING IN THE 3C4 REALIZATION PHASE

After the concept phase, the target costing models became more detailed. The current report Tim prepared for the 3C4 project team is shown in Table 2. It may not look much more granular than Table 1, but that is because it shows only the aggregated numbers. Tim had written a short memo explaining this report to SE team members (see Exhibit 2).

While preparing for the review meeting of the 3C4 project team, Tim went through his recollections of important things that had happened during the realization phase of the 3C4 project. From the beginning, the challenging nature of this project had been apparent. For the first time, BMA would be offering a plug-in hybrid and full-electric drive for the 3C4, which made the model much more complex than the previous generation. While this complexity had been considered in the final plan after the concept phase, BMA had less experience with these technologies, and during the realization phase, BMA was encountering more unexpected events. For example, Tim knew that technical issues with regenerative braking was one reason for the materials costs to be above target. Because of such problems, more R&D had been done to find technical solutions, which had raised development costs. The complexity of the 3C4 also meant more outsourcing of development to suppliers and larger investments with suppliers, even beyond what had already been expected at the end of the concept phase.

Another development was that BMA expected competition to become tougher in a large number of countries. In response to that, BMA had decided to upgrade some specifications but without adjusting expected sales prices. For example, the base configuration in many countries now included leather seats. Also, the target costing model for the realization phase included more detailed and updated information about sales prices, reflecting an unfavorable trend in comparison to the earlier, less detailed pricing assumptions for the reporting models. Moreover, the exchange rates of the euro against the dollar and the renminbi had changed in unfavorable directions.

As if all this were not enough, in its review of a prototype the BMA's board of directors had demanded a "minor adjustment" in the interior of the car. This adjustment,

however, meant that the center console had to sit 4 millimeters lower, with implications for several parts of the body and instrument panels. The changes did not increase production costs but caused another cost overrun in development.

### THE NEXT 3C4 PROJECT TEAM MEETING

The next meeting of the 3C4 project team would happen in two days. As the project was halfway into the realization phase, the purpose of the meeting was to review the current status of the project and to decide on actions, because not everything was going according to plan.

Tim decided to send a cost-reduction proposal to project team members. He wondered how the representatives from the various functions (such as engineering, procurement, production, and marketing) would react and which other ideas they might bring forward during the 3C4 project review meeting.

Furthermore, the 3C4 project team manager had asked Tim to explain BMA's target-costing system during the next meeting. Several new members had recently joined the project team, two of whom were from outside BMA. They had previously worked at KeyHole Schloss GmbH and had little experience in automotive manufacturing. Tim had received a presentation of target costing at KeyHole Schloss (see Appendix C). He realized that target costing was different and more intricate at BMA than at KeyHole Schloss, and he wondered how to bring these new colleagues up to speed with target costing at BMA.

## PART 2: EXPANDING BMA'S TARGET COSTING METHOD

The experiences with the 3C4 model were illustrative of how managing costs had become much more challenging. Laura Forsbeck, BMA's new CFO, wanted to change BMA's target-costing system for future projects. Almost two years after Part 1 of the case (several months after the Start of Production milestone for the 3C4 model), Laura had formed a task force, led by Tim, charged with the task to redesign BMA's target-costing system.

### LIMITATIONS OF BMA'S TARGET-COSTING METHOD

Laura considered BMA's current target-costing system as inadequate, because it basically neglected costs for R&D and manufacturing equipment. The initial investments for R&D and manufacturing equipment were a major cost category (see Tables 1 and 2). The logic for calculating these cost targets, however, was rather reactive and not market-driven. For example, the cost targets for the new 3C4 had been derived from the actual costs of the previous generation of the 3C4. Tim and other target-costing experts had extrapolated those actual costs by making assumptions about cost savings that would be possible owing to technological progress and about extra investments that would be needed, particularly for electric drives. Overall, much higher costs were expected for the new generation of the 3C4 model, and cost targets had set been considerably higher. Laura believed it should be possible to budget the costs for R&D and manufacturing equipment in another way, more in line with how material costs were managed in the target-costing logic.

Another limitation of BMA's current target-costing system Laura saw was that it considered an overly narrow part of the product portfolio, such as only the 3C4 model. The system did not focus enough on possibilities for reducing costs across BMA's car models. She believed cost management required better coordination of design decisions across models. Parts sharing was a key example. Obviously, various car models already used the same engines. Developing and manufacturing different engines for each model would be very inefficient and unnecessary. Some other components were also shared, but Laura believed many more possibilities existed to share parts and save costs in development and manufacturing. She also realized that while sharing parts could be beneficial, that did not always have to be the case. Tradeoffs mattered for either developing a new, model-specific part or sharing an existing, common part. Laura showed a slide that listed some of the potential cost savings and possible cost increases due to parts sharing. The new target-costing system should stimulate better decisions about sharing or not sharing components, which would optimize costs in development, purchasing, and manufacturing for BMA as a whole.

### COSTS OF EXISTING (SHARED) VS. NEW (UNIQUE) PARTS

Tim considered how the current target-costing system worked for various kinds of parts. When designing a new car, the engineers employed a mix of existing parts that BMA already used for current cars and new parts developed specifically for a new car model. Project managers responsible for the development of a new car were completely free to make their own decisions about using an existing part or developing a new part.[2] Project managers would usually consider both options for many parts and review the cost differences for their own project. In particular, the items material costs, development costs, and manufacturing investments in BMA's target-costing system were relevant (see Table 2).

In the "Material Costs" line under the "Actuals" column, for existing externally sourced parts, costs were based on the current purchase prices. For internally produced parts, costs were based on the current full manufacturing unit costs. For new parts, the practice was basically the same, but those costs needed to be based on estimates of the purchase prices or the full manufacturing costs.

In the "Development Costs" line under the "Actuals" column and in the "Manufacturing Investments" line under the "Actuals" column, for existing parts, costs were shared among the car models that used a part, with costs based on the number of parts used by each model. For example, suppose the costs for product development and initial investments had been €35 million for a particular part and now another car model also started using this part, using about 20% of the total number of units of that part. The new car would then be charged €7 million, and this amount would be credited to the other car models that were already using this part. For new parts that were specially developed for a particular new car model, the project was charged the actual development costs and initial investments for those parts.

### THE TASK FORCE KICKOFF MEETING

The first meeting of the task force is in three days, and Tim is preparing his presentation. He wondered how he would explain the challenge and task to the task force.

Tim shared Laura's doubts about how BMA approached the costs for R&D and manufacturing equipment. He believed they should develop an approach for how BMA could deal with the R&D and manufacturing equipment costs differently.

Tim agreed with Laura's emphasis on better decisions on parts sharing. He realized that sometimes having a common part was optimal, but in other cases having a unique part was preferable—it was a complex tradeoff. He wondered which potential cost savings and possible cost increases owing to parts sharing Laura might expect. He also wondered if the current target-costing system created incentives for individual project managers to make decisions about sharing (or not sharing) of parts that would not be in the best interest of BMA. He believed the task force should develop ideas for adjusting the target-costing system so that parts sharing would be improved from BMA's point of view—that is, for the entire portfolio of car models.

## ENDNOTES

[1]Later, the product description document described the chosen technical solutions for realizing the customer requirements. These technical aspects included, for example, engine power and torque, car weight, the drag coefficient (Cd value), car body materials, wheel size, and types of axles, suspension, and brakes.

[2]In principle, several managers responsible for developing a number of new car models could also jointly decide to start developing a new part they would share. Costs for product development and initial investments would then be allocated to these car models in the same way as for existing parts—based on volume. But such joint development projects almost never occurred.

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## APPENDIX A. BMA'S PRODUCT-DEVELOPMENT PROCESS

This appendix provides background information on BMA's product development process, which consists of two phases: the concept phase for product definition and the realization phase. These phases were broken into 15 milestones (see Exhibit). The project organization for product development consisted of several different teams.

### PROJECT MILESTONES

Major milestones regarding the market include market positioning (such as an analysis of competitor models and a review of the current model), product profiling (such as a definition of the customer requirements for the new car model), initial targets for volumes and sales prices, and commitments to detailed targets for volumes and sales prices.

Major milestones regarding finance include setting the target profit for the project, formulating top-down targets for major areas (such as the powertrain and electronics) and functions (such as procurement and quality management), committing to these high-level targets and to detailed targets at the parts level, and verification of the business case.

Major milestones regarding design (visualization) include creating the initial design brief and making subsequent design decisions based on representations that steadily became more realistic (going from 2-dimensional to 3-dimensional images that are more photo-realistic and producing physical models—first of the exterior and later of the interior), and finally freezing the design.

Milestones regarding the product include advancing the car's technical design. Initially, specific decisions defined the technical properties of the car (they were described in the product description document) leading to the so-called concept freeze. Further milestones concerned the technical realization of the car through the development of the major parts and the design of the car body, and subsequent release of the car in various stages.

Milestones regarding purchasing include selecting development partners for concept design and asking for quotes, selecting suppliers, and signing contracts.

### PROJECT ORGANIZATION

Parts are developed by simultaneous engineering teams (simultaneous engineering teams). Typically, about 35 simultaneous engineering teams are needed for parts such as the dashboard, exterior lighting, interior lighting, door panels, and seats. A simultaneous engineering team comprised members from engineering, procurement, production, quality, finance, marketing, and sales.

The next level in the project organization consisted of the technical module teams, each of which was responsible for a major area of the car:

- Powertrain (engine, transmission)
- Chassis sub-assemblies (axles, suspension, brakes, wheels)
- Car body[1]
- Fittings (such as seats, arm rests, interior panels, and dashboard)
- Electronics

**Exhibit. BMA's Product Development Process (Selection of Milestones)**

| | Concept Phase: Product Definition | | | | | Realization Phase: Product Development | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Project Start | Product Mission | Project Pre-suppositions | Product Definition | Project Feasibility | Product Concept Decision | Product Design Decision | Design Freeze | Procure-ment Release | Product Launch Release | Release of Pre-Production Cars | Release of Trial Production | Pre-production Series | Start of Production | Market Entry |
| Weeks | -54 | -48 | -45 | -42 | -39 | -33 | -27 | -23 | -18 | -12 | -8 | -6 | -3 | 0 | 3 |
| **Market** | | Market positioning | Product profiling | Initial volume and price targets | Detailed volume and price commitments | | | | | | | | | | |
| **Finance** | | | Project profitability target | Top-down cost targets | Target commitment | Target commitment at part level | Business case veri-fication | | | | | | | | |
| **Design** | | Initial design brief | | | Subsequent design decisions | | | Design freeze | | | | | | | |
| **Product** | | | | Product definition | | Concept freeze | In various stages: technical development of the major parts and technical design of the car body, release of various series | | | | | | | | |
| **Purchasing** | | | | | Selection of development partners | | In various stages: asking for quatations, supplier selection, and contracting | | | | | | | | |

[1]The car body is what mainly appears from the outside as "the car." An inside structure determines the shape and strength of the car and the outside panels are what the public sees. The lower part of the body, also called the "platform," is where the powertrain, chassis subassemblies, and seats are connected. Not much of this is visible from the outside. The upper part of the body—consisting of, for example, the side frames, pillars, and roof— determines the visual appearance of the car, such as the height, number of doors, and silhouette. Most of what is visible composes the attached parts, such as the hood, fender, doors, trunk lid, and bumpers. These are usually connected to the body with screws, often with a hinge.

The project team was responsible for the entire project for developing a particular new model. Next to the project leader, other members were the leaders of the technical module teams and a high-level representative from each of the various functional departments of engineering, procurement, production, quality, finance, and marketing and sales.

After sometimes-intense discussions, most decisions were made by the simultaneous engineering teams. Some decisions needed to be made by higher-level teams in the project organization. For example, the executive board of directors made all decisions about design, as well as decisions about car connectivity, that were needed to enable new business models. Also, thresholds were used regarding the financial implications of a decision. These thresholds specified when a decision had to be made by a technical module team, the project team, or even BMA's executive board of directors. Decisions in the simultaneous engineering teams were based on voting. But if one function raised strong objections against the majority decision, that decision would be escalated to the next level.

## APPENDIX B. SEVERAL DECISIONS IN THE 3C4 CONCEPT PHASE

Tim recalled several technical concept decisions that were heatedly debated during concept development of the new 3C4 model.

The material of the car body. The body could be built largely with aluminum or steel. Aluminum would be about 110 kilograms lighter, which favorably affected handling, fuel economy, and emissions, but was also more expensive. Steel was stronger and easier to repair. The engineering experts had estimated the cost difference to be around €355 per car. Over the total lifecycle of the car, this would amount to hundreds of millions of euro, which made this a board-of-directors decision. At the end, the decision was to use a mix of materials and to optimize which parts would be made of what material. The result was a clever design of steel and aluminum parts that achieved a good weight distribution. Compared with steel only, it reduced the weight of the car by about 70 kilograms and limited the cost increase to €160 per car.

**Automatic transmission.** Given the market positioning of the new 3C4 model, marketing required that it should always come with an automatic eight-speed transmission. The cost impact estimate projected that having only this automatic transmission saved considerable development costs for a new manual transmission. The unit manufacturing cost of the automatic transmission was more expensive. But some manufacturing costs could be saved if only one type of transmission needed to fit in. On balance, offering all cars with an automatic eight-speed transmission would increase the cost per car by about €275, which was seen as justifiable given the positioning and pricing of the new 3C4 model.

**Dashboard design concept.** The product profile document required a dashboard with more elegant and expensive-looking materials and a more interesting design. The dashboard of the current 3C4 was based on the one-part concept.[1] The new 3C4 would adopt the two-part design concept, with a horizontal top part and a separate, vertical part below it (facing the passengers) that resulted in a more luxurious and interesting design. For example, customers could choose different types of materials for the two parts, ambient lighting could be integrated into the dashboard, and a more intricate design with more depth was possible. Yet, the two-part concept involved higher costs of about €112 per car, and the target costing model showed that these higher costs were problematic. Further discussions that also involved top management led to adoption of a two-part concept, and cost targets for some other parts were subsequently reduced.

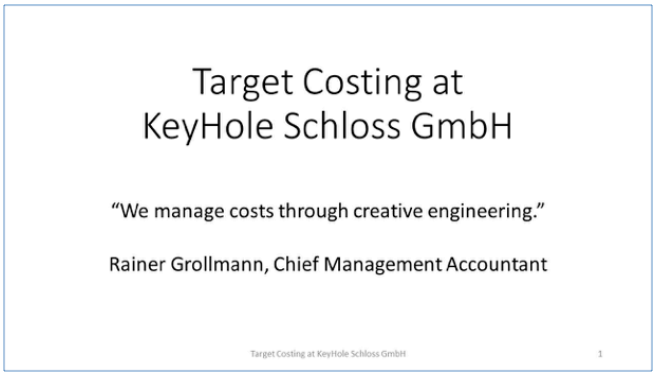
**Brakes.** The requirements for the new car also included excellent braking performance that would help differentiate the driving experience of the new 3C4 model from some of its competitors. These requirements could only be realized by choosing a fixed-caliper disk brake, as opposed to a floating-caliper disk brake that almost all other cars in this segment used. The extra costs were estimated to be about €178 per car. This cost increase created a lot of discussion, because the allowable cost for the brakes did not have any room for an increase. Tim had argued that it should be possible to achieve the desired braking performance another way. The discussion needed to be escalated, and in the end, the fixed-caliper disk brake was not adopted in the technical concept.

**Suspension.** The requirements also demanded excellent suspension that would provide a best-in-class comfortable ride. Engineering had proposed the use of air suspension in place of conventional steel springs for all cars of the new 3C4 model. This proposal also created much discussion, because the target costs for the suspension did not allow for air suspension. Yet, the decision was that this item would be an important distinguishing feature of the new 3C4 model and needed to be implemented. Some cost reductions had to be realized for other chassis subassemblies. Fortunately, some cost reductions would be facilitated by BMA's purchase of air suspension in a far greater volume than thus far.

[1]The one-part concept means that main part of the dashboard starts horizontally behind the windshield and curves toward the passenger (vertically) to end at the glove compartment. The dashboard is made of one piece and one type of material.

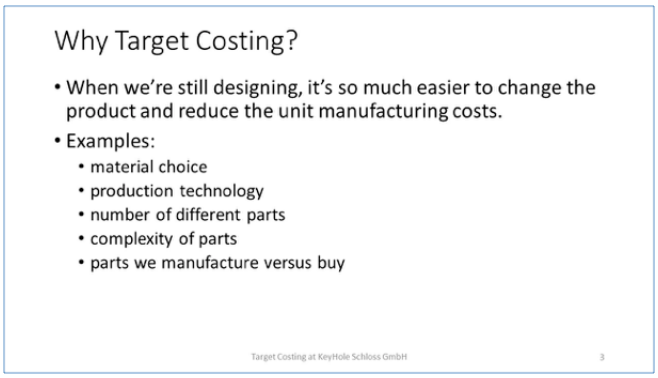
**Six-cylinder engine.** Should the new 3C4 have the option for a six-cylinder engine? A new six-cylinder engine was available that had been developed for other cars. Recent market research had indicated that a limited number of customers were expected to choose this engine, but they would be willing to pay a considerable price premium.[1] However, having this option would affect the design of the body of all cars (with or without a six-cylinder engine) and slightly increase the manufacturing cost of all units. The financial analysis required a business case that considered the incremental costs and revenues. This business case for optionality for a six-cylinder engine was positive and this option was included in the technical concept.

## APPENDIX C. PRESENTATION ON TARGET COSTING AT KEYHOLE SCHLOSS



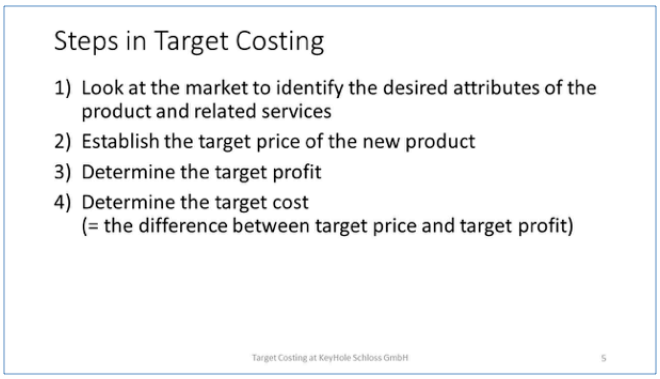


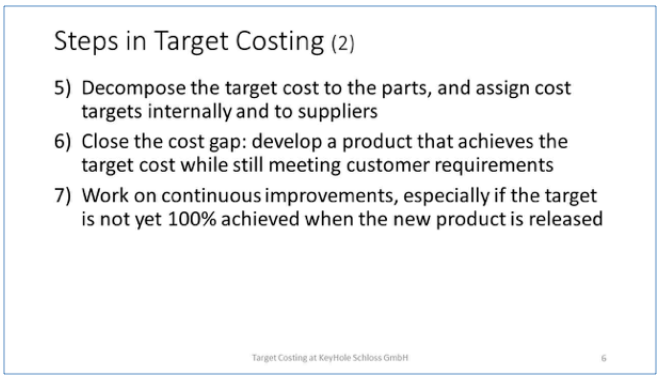


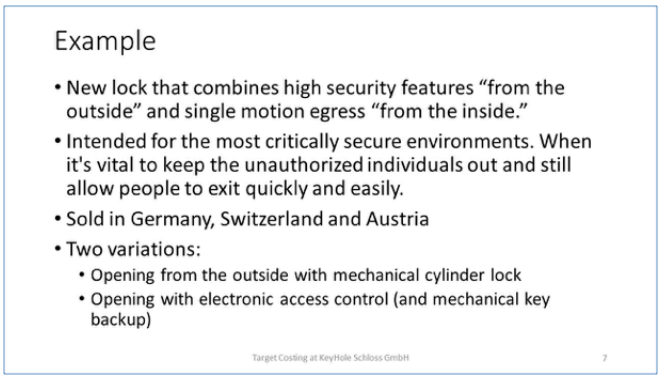









[1]To support decisions about pricing of the base configuration and options, BMA had detailed data per country and per year for how sales prices were affected by attributes of the car. For example, BMA could estimate how much customers would pay for fuel economy (in euros per increase of kilometer/liter), engine power (euros per kilowatt), more cylinders, or an automatic transmission.

## Example (2)

- Market research had indicated that prices of around €450 and €650 would be reasonable for customers.
- We expected mix of 60%/40%, giving a weighted average sales price of €530.
- Distributors require a margin of around 40%, we require a gross margin of 45%, so the target cost was €175.

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## Example (3)

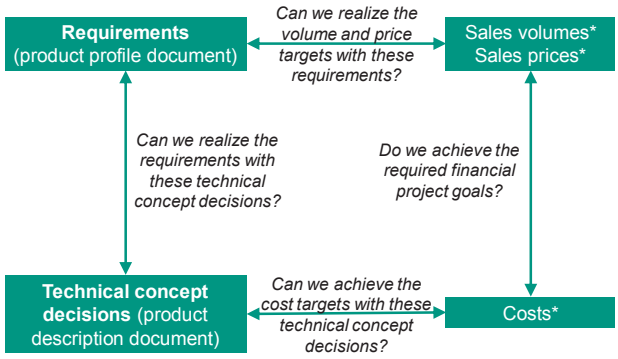
- Initial designs had a unit cost of around €210.
- We worked on the design and achieved the target cost through
  - more selective use of very expensive materials,
  - redesigning parts so they could be produced more easily, and
  - cancelling the requirement to be bullet-proof (only needed for very few customers).
- Took around 6 months, as planned

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**TABLE 1. EARLY TARGET COSTING REPORT**
(for one of the reporting models at a given point in time during the concept phase)

| Early Target Costing report<br>Reporting model: Germany | |
|---|---|
| Sales Price | €38.250 |
| Local Sales Tax | €6.108 |
| Dealer Margin | €5.142 |
| Net Revenues | €27.000 |
| Material | €12.750 |
| Direct Labor | €1.600 |
| Warranty | €900 |
| Sales Expenses | €900 |
| Logistics | €750 |
| Contribution Margin | €10.100 |
| Initial Investments | €3.300 |
| Overhead Costs | €3.500 |
| Risk Surcharge | €600 |
| Operational Profit | €2.700 |
| as Percentage | 10,00% |

**EXHIBIT 1. TARGET-COSTING LOGIC DURING CONCEPT DEVELOPMENT**




** Target costing analyses based on reporting models, as shown in Exhibit 1.*

**TABLE 2. FULLY DEVELOPED TARGET-COSTING REPORT**

## Target Costing Report

in euro per car

| | Targets | Previous Report | Actuals | Actuals – Targets |
|---|---|---|---|---|
| | Nov 2016 | Aug 2017 | Nov 2017 | better/*worse* |
| Life Cycle Volume (cars) | 260.000 | 260.000 | 265.000 | 5.000 |
| **Net Revenues** | **27.271** | **27.050** | **26.839** | *-432* |
| Material Costs | 13.450 | 13.600 | 13.800 | *350* |
| Direct Labor and Machine Costs | 902 | 900 | 886 | -16 |
| Warranty Costs | 410 | 408 | 405 | -5 |
| Sales Costs | 396 | 395 | 390 | -6 |
| Logistics | 497 | 495 | 491 | -6 |
| **Total variable costs** | **15.655** | **15.798** | **15.972** | *317* |
| (as % of net revenues) | 57,4% | 58,4% | 59,5% | *2,1%* |
| **Contribution Margin** | **11.616** | **11.252** | **10.867** | *-749* |
| (as % of net revenues) | 42,6% | 41,6% | 40,5% | -2,1% |
| Development Costs | 2.495 | 2.543 | 2.643 | *148* |
| Manufacturing Investments | 1.715 | 1.797 | 1.852 | *137* |
| Marketing | 235 | 235 | 235 | 0 |
| Startup Costs | 179 | 180 | 180 | *1* |
| Total initial costs | **4.624** | **4.755** | **4.910** | *286* |
| (as % of net revenues) | 17,0% | 17,6% | 18,3% | *1,3%* |
| Indirect Manufacturing Costs | 2.170 | 2.196 | 2.221 | *51* |
| Procurement | 520 | 528 | 540 | *20* |
| General Management | 1.163 | 1.172 | 1.198 | *35* |
| Risk Management | 400 | 400 | 400 | 0 |
| Total Overhead Costs | **4.253** | **4.296** | **4.359** | *106* |
| (as % of net revenues) | 15,6% | 15,9% | 16,2% | *0,6%* |
| **Profit** | **2.739** | **2.201** | **1.598** | *-1.141* |
| (as % of net revenues) | 10,0% | 8,1% | 6,0% | *-4,1%* |
| Project NPV (Million €) | 23,8 | -9,1 | -63,8 | *-87,6* |
| Project IRR (in %) | 9,7 | 7,8 | 4,6 | *-5,1* |

Note: All monetary values are represented in euros.

**EXHIBIT 2: MEMO EXPLAINING THE TARGET COSTING REPORT**

**THE TARGET COSTING REPORT**

**Targets column:** Contains the targets set for the 3C4 model at the end of the product concept phase as the basis for the business case and the "go" decision.

**Actuals column:** Currently (at the reporting date) the most likely estimate.

**Previous Report column:** Shows the Actuals from the previously issued report.

**Life Cycle Volume:** Based on detailed sales volume targets: for six years, six powertrains (combination of engine + transmission + 2 or 4 wheel drive) and 120 countries.

**Net Revenues:** "Net" means after subtracting sales taxes, dealer margins, and importer margins (estimated per country). Sales price targets are defined per country, year and powertrain, using country-specific base configurations. Each country also estimates an average revenue per car for the additional options customers are likely to buy.

**Materials Costs:** Material costs are built up from targets and estimates for about 2000 different car parts. The base configurations are an input for estimating the materials costs. Material costs for options are estimated as a percentage of the revenues for options.

**Direct Labor and Machine Costs:** These costs are based on the production times and the hourly rates for labor and machines in the various product departments, such as the press shop, body shop, painting, assembly, and engine production.

**Warranty, Sales, Logistics Costs:** The report is based on country-specific targets for these costs, as well as updated estimates (Actuals).

For the various cost categories mentioned above, the target costing model includes separate costs targets for each of the six years. Cost reduction after start of production is required, because sales prices go down during the product life cycle.

The report includes another panel for the initial costs (listed below) that are direct in relation to the 3C4 model and fixed relative to volume. These costs are estimated in total and divided by the total volume, and are shown as a per unit cost number..

**Development Costs:** The target is based on the detailed project plan defined at the end of the concept phase. This includes the product upgrade, about halfway into the product life cycle. The Actuals are based on adjustments of the initial project plan and new estimates.

**Manufacturing Investments:** See Development Costs.

**Startup Costs:** These costs relate to the time for training production employees, special tools for that purpose (not used later in production) plus materials. Also included are external training costs, plus costs of lost production when the line is used for training and for building prototypes and early series.

The final panel includes indirect costs that are allocated to the 3C4 model. These are outside the control of the 3C4 model and are included to get a more comprehensive profit calculation.

Production times are used for allocating the fixed **Indirect Manufacturing Costs** to the 3C4 model. These indirect costs per hour are very diverse; illustrative numbers are €500 for presses and €30 per labor hour in assembly.

**Procurement** costs (inbound logistics, warehousing, supplier management, etc.) are allocated via a surcharge on purchase prices.

**General Management** (including many indirect departments), and

**Risk Management** costs (basically a buffer for unexpected costs) are allocated via surcharges on variable manufacturing costs.